AERO 4442A: Trajectory Assignment

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INTRODUCTION

The exploration of space and the deployment of payloads into specific orbits have long been facilitated using sounding rockets. These vehicles provide a unique platform for scientific research, technology testing, and educational purposes. The performance of a sounding rocket is fundamentally characterized by its ability to reach a desired apogee and achieve a specific range, parameters that are influenced by the rocket's design, propulsion system, and the external forces acting upon it during flight.

This report aims to construct a comprehensive performance map that delineates the relationship between the apogee and range of a specified sounding rocket, considering varying launch angles and payload masses. The significance of this analysis lies in its potential to inform design considerations, optimize performance, and enhance the payload delivery capabilities of future sounding rockets. By systematically evaluating the impact of aerodynamic drag and other forces throughout the rocket's trajectory, this study seeks to provide insights into the methods used to complete this simulation.

NOMENCLATURE

V ..... Velocity (m/s)
a_g ..... Acceleration due to gravity (m/s^2)
T ..... Thrust (N)
T ..... Temperature (Kelvin)
D ..... Drag force (N)
ρ ..... Air density (kg/m^3)
S_{ref} ..... Reference area of the rocket (m^2)
C_D ..... Drag coefficient
θ ..... Launch angle (degrees)
M ..... Mach number
Δt ..... Time step (s)
u_{eq} ..... Exit Velocity (m/s)
m ..... Mass (kg)
γ ..... Ratio of specific heats
R ..... Universal gas Constant

METHODOLOGY

This study employs a numerical simulation approach to evaluate the performance characteristics of a sounding rocket, specifically focusing on its apogee and range under varying launch angles and dry mass conditions. The simulation is implemented in MATLAB, leveraging parallel computing capabilities to expedite the analysis across multiple scenarios. The methodology used and assumptions made are described below.

Atmospheric Model

The atmospheric conditions encountered by a rocket during its ascent significantly influence its trajectory and overall performance. These conditions, including temperature, air density, speed of sound, and gravitational acceleration, vary with altitude and are critical for accurate trajectory modeling.

For this analysis, the 1976 US Standard Atmosphere model [1] serves as the foundation for simulating atmospheric conditions. This model provides detailed profiles for air density, temperature, and gravity acceleration up to an altitude of 1000 km. Plotted below is the density and acceleration of gravity as a function of the altitude.

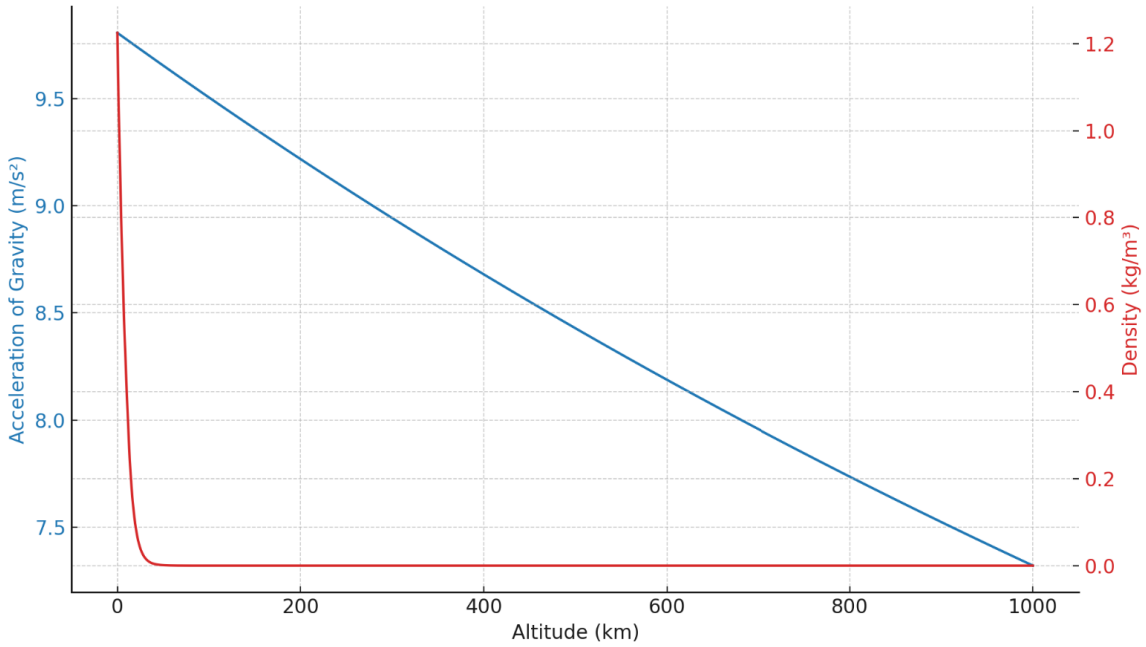


Figure 1: Atmospheric Model: Density and Acceleration of Gravity

The model delivers direct values for the speed of sound up to an altitude of 86 km, above this threshold, the speed of sound is determined using the following equation:

$$a = \sqrt{\gamma RT} \tag{1}$$

The atmospheric model used provides the speed of sound up to 86 km. After this, we use the model temperature to find the speed of sound. The figure below depicts the speed of sound as a function of altitude used for this simulation.

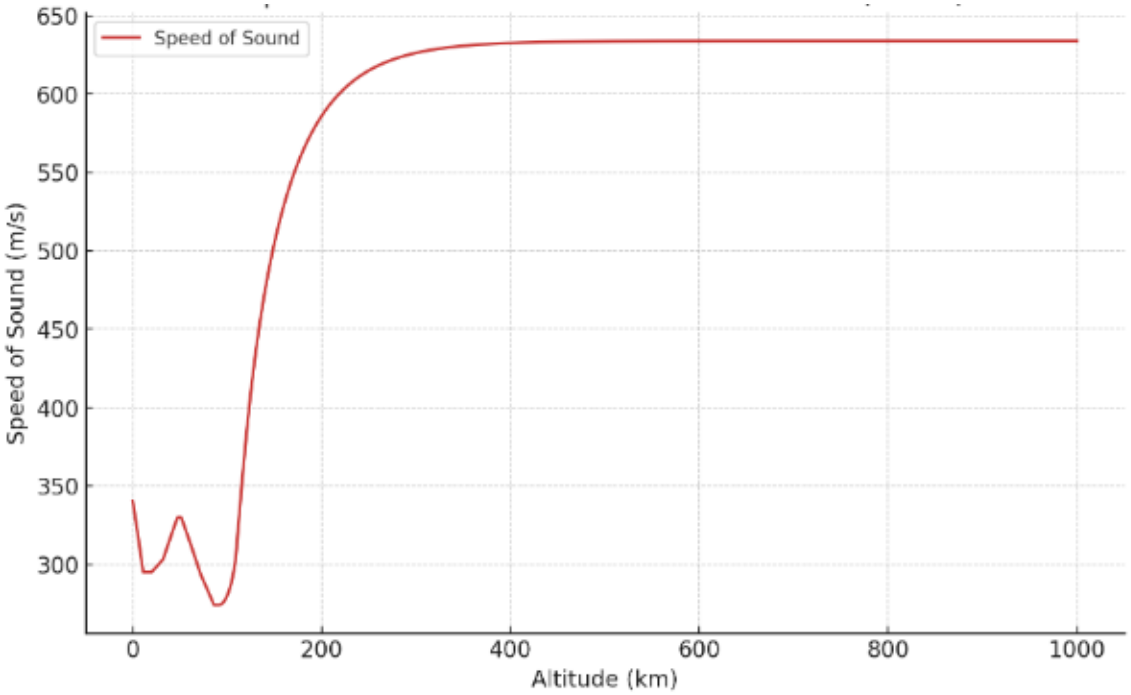


Figure 2: Atmospheric Model: Speed of Sound

Dynamics

The dictating equation used to model the rocket's dynamics, as outlined in the course text (equation 4.1), is referenced here as Equation 2:

$$\frac{\partial}{\partial t} mV = m_{pr} u_{eq} - D - m a_g \cos\theta \quad (2)$$

Equation 2 can be rearranged to define the acceleration, velocity, and position of the rocket throughout its trajectory. The x and y components of the rocket's acceleration at a given time are shown below in Equations 3 and 4:

$$a_x(t) = \frac{T_t \sin\theta_t - D_t \sin\theta_t}{m_t} \quad (3)$$

$$a_y(t) = \frac{T_t \sin\theta_t - D_t \sin\theta_t - a_g y}{m_t} \quad (4)$$

Note that the mass at any time step is a function of the thrust and ISP of the rocket. The exit velocity can be found by.

The exit velocity is given by:

$$u_{eq} = ISP \cdot a_g \quad (5)$$

Using this exit velocity and thrust, the mass flow rate of propellant can be found as:

$$\dot{m}_{pr,t} = \frac{T_t}{u_{eq}} \tag{6}$$

The mass at any timestep can be found as:

$$m_t = m_{t-1} - \dot{m}_{pr,t} \tag{7}$$

The initial mass of the rocket is:

$$m_{\text{initial}} = m_{\text{motor}} + m_{\text{payload}} + m_{\text{fuel}} \tag{8}$$

Given the acceleration of the rocket at any given timestep, we can find the velocity and position of the rocket in a similar fashion, shown below in Equations 9 and 10 for velocity, and 11 and 12 for the position:

$$V_{x(t)} = V_{x(t-1)} + a_{x(t-1)}\Delta t \tag{9}$$

$$V_{y(t)} = V_{y(t-1)} + a_{y(t-1)}\Delta t \tag{10}$$

$$x_{(t)} = x_{(t-1)} + V_{x(t)}\Delta t \tag{11}$$

$$y_{(t)} = y_{(t-1)} + V_{y(t)}\Delta t \tag{12}$$

Note that the reference frame for this report was chosen to have an origin at the base of the launch rail.

Gravity Turn

After the rocket leaves the launch rail, gravity turn will start to act. This results in a change in the rocket's acceleration in the x and y axes. This small change of angle will have a significant impact on the trajectory of a rocket throughout its flight. The equation represents (equation 4.1.1 in course text [2]). The rate of change of this angle is shown below in Equation 13:

$$\theta_{(t)} = \frac{a_g \sin\theta_{(t-1)}}{V_{(t-1)}}\Delta t \tag{13}$$

Thrust

The thrust table provided in the assignment package [3] was integrated into the code, and at each timestep, the thrust was interpolated from this "lookup" table. The thrust of the rocket throughout the flight is shown in the results section of this report.

Drag Force

The drag acting on the rocket is a function of the geometry, the speed the rocket is traveling, and the medium it is traveling through. The drag force can be described by the following equation:

$$D = 0.5 \cdot C_D \cdot V^2 \cdot \rho \cdot S_{\text{ref}} \tag{14}$$

Note that S_{ref} is a function of the cross-sectional area of the rocket, as outlined in the assignment package [3].

The Mach number can be defined as:

$$M = \frac{V}{a} \tag{15}$$

Once the Mach number is known, we can determine the coefficient of drag (C_D). (C_D) can be defined as Equation 16

$$C_D = C_{wave} + C_{skin} + C_{base} \tag{16}$$

As the Mach number increases, the equations used to find C_{wave}, C_{skin}, and C_{base} change. This is incorporated into the model as conditional statements that are functions of Mach number M.

Wave Drag

$$C_{D_{\text{wave}}} = \begin{cases} 0.06157 \exp(1.4075M) & \text{if } M \leq 1.2, \\ 0.40919 M^{-0.3405} & \text{if } M > 1.2. \end{cases}$$

Base Drag

$$C_{D_{\text{base}}} = \begin{cases} 0 & \text{if } M \leq 1.2, \\ 0.26077 M^{-1.00589} & \text{if } M > 1.2. \end{cases}$$

Skin Friction Drag

$$C_{D_{\text{skin}}} = \begin{cases} -(0.072158 \log(M)) + 0.183106 & \text{if } 0 < M \leq 9, \\ 0 & \text{otherwise.} \end{cases}$$

MATLAB Model

MATLAB was used to to implement the equations and assumptions discussed above.

Atmospheric conditions and drag coefficients are determined using interpolation functions based on altitude. The simulation loop iterates as long as the rocket's altitude is above ground, updating time, velocity, position. The simulation recalculates the rocket's velocities both horizontal and vertical using the acceleration data from the previous time-step (0.01 seconds). The rocket's position is then adjusted accordingly.

The model then updates the atmospheric density and the speed of sound. These adjustments are based on the rocket's altitude, and the atmospheric model used.

The engine's thrust is derived from its specific thrust profile, which takes into account the duration of propellant burn. Simultaneously, the function calculates the drag force acting against the rocket. This drag is contingent upon the rocket's velocity, density, and the drag coefficient that varies with the Mach number.

Throughout the burn time, the rocket's mass decreases over time. The simulation updates this mass until burnout after burnout the model assumes the mass of the rocket to stay constant. Then finally the accelerations in both the horizontal and vertical planes, considering the cumulative effects of thrust, drag, and gravity.

RESULTS

The apogee elevation and impact range of the rocket are compared to the results from the Black Brant V rocket. The launch angles vary from 5-15 degrees from the vertical, and the payload masses considered are shown in Table 1.

Table 1: Payload Masses

Payload Mass (kg)
226.8
272.2
317.5
362.9
408.2
453.6
499.0
544.3

It was found that the values provided in the assignment package likely corresponded to the Black Brant V rocket [4]. Performance charts for this rocket are used to validate the results of this report. Shown below in Figure 1 every combination of launch angle and payload was modeled and compared to the results published in [cite].

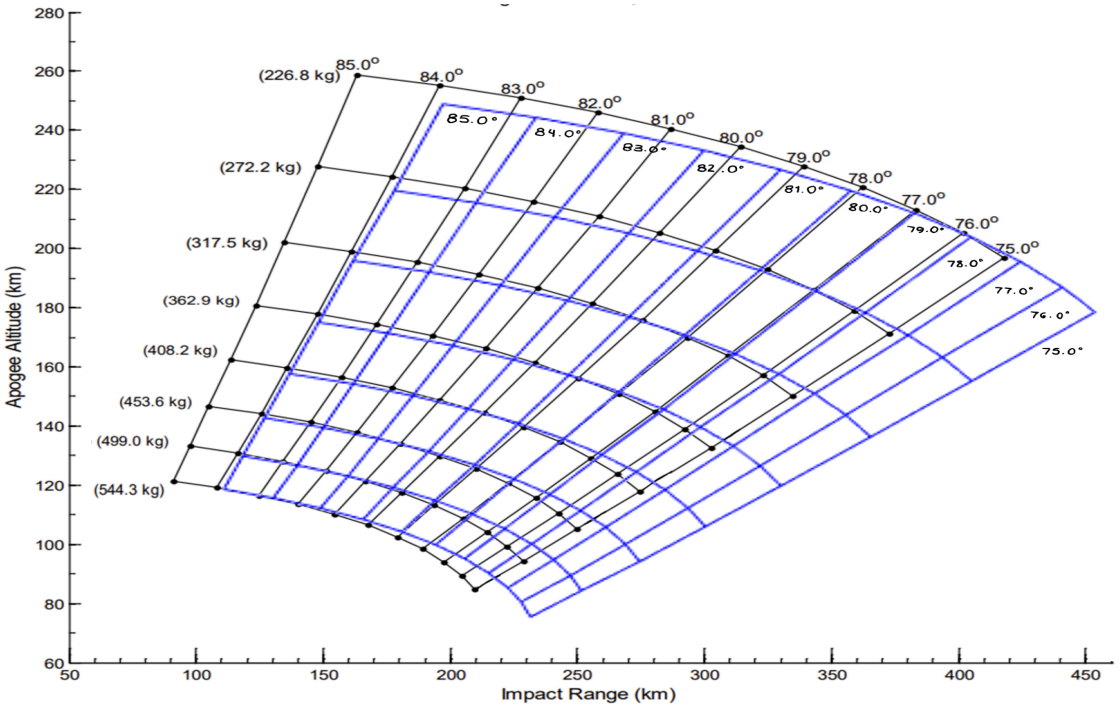


Figure 3: Performance Comparison With Black Brant V

Please note that in Figure 1 the angles are from the horizontal. It Can be seen that there are dissimilarities in the results. The simulated results correspond to a launch angle of 85 degrees from the horizontal fall in line with the expected results at a launch angle of 84 degrees. Additionally,

across all scenarios, the simulations predict a greater range and lower altitudes at apogee. This deviation might be the thrust values used, with interpolated thrust not matching the rocket's actual thrust at specific times. Another potential source of this error might stem from the implementation of the gravity turn in the model. Furthermore, the length of the launch rail utilized in the model, which is 19 meters, could contribute to this error. Once the rocket surpasses the launch rail, the gravity turn begins to influence its trajectory. A longer launch rail would likely lead to higher apogee values and shorter impact ranges for the rocket.

Figure 4 below shows the flight profiles for the minimum payload at the minimum launch angle and the maximum payload.

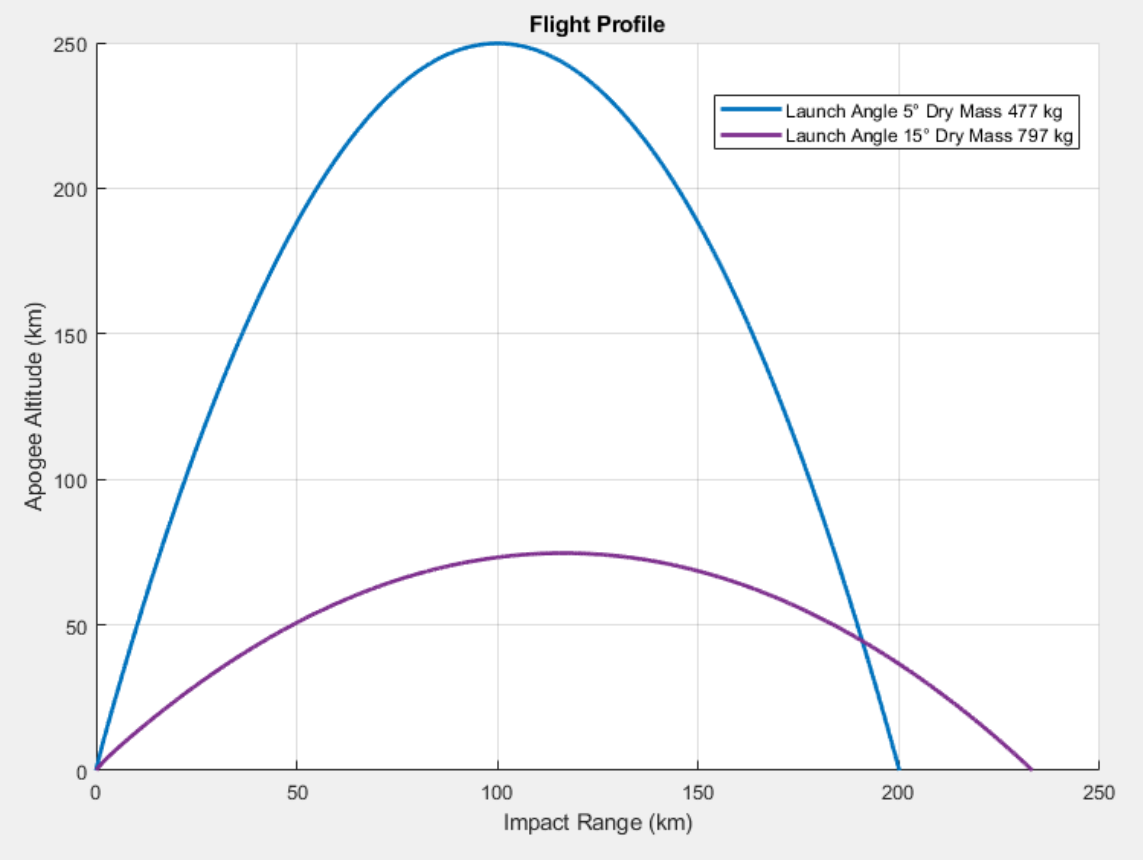


Figure 4: Flight Profile

These scenarios depict the maximum and minimum apogee and range modeled. When simulating a trajectory with a launch angle of 5 degrees and a dry mass of 477 kg (including a payload of 225 kg), the rocket attains an apogee of 248 km and a range of 198 km. With a launch angle of 15 degrees and a dry mass of 797 kg (carrying a payload of 545 kg), the rocket reaches an apogee of 74 km and an impact range of 233 km. This constitutes a 70 percent reduction in apogee and a 17 percent increase in impact range.

The figure below shows a performance chart that compares the outcomes with and without considering drag. When drag is neglected, each scenario achieves higher apogee altitudes and longer impact ranges, as anticipated.

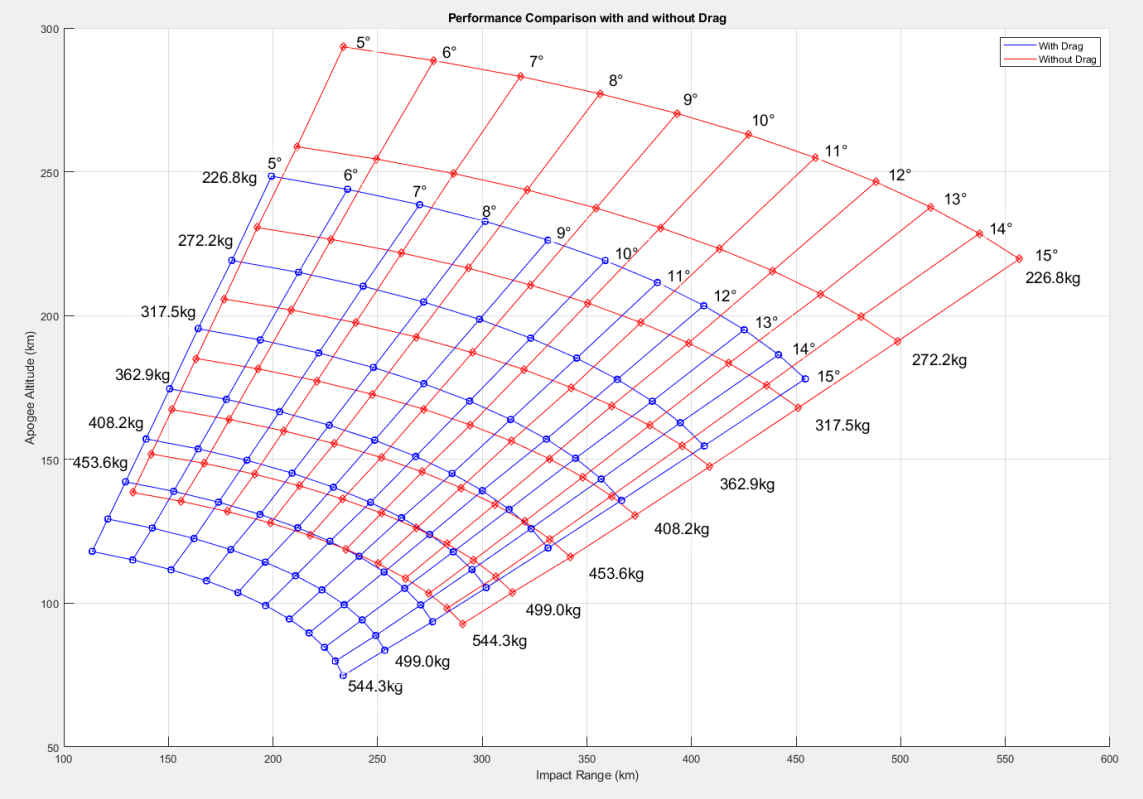


Figure 5: Performance Comparison with and without Drag

For the lightest payload at a 5° angle, the decreases in apogee and impact range are approximately 15.36% and 14.59%, respectively. For the heavy payload at the same angle, these reductions are around 15.22% and 14.39%, respectively. At a 15° launch angle, the lightest payload experiences a decrease in apogee and impact range by about 18.72% and 18.35%, respectively, while for the heavy payload, the decreases are approximately 19.57% and 19.66%, respectively. These results are consistent with the expectation that drag has a more pronounced impact on scenarios with larger launch angles, due to the trajectory spending more time in denser parts of the atmosphere. Furthermore, drag significantly influences lighter payloads due to its dependence on the velocity squared; thus, a lighter payload, moving at higher speeds, experiences a more substantial impact from drag.

Further analysis reveals the effects of these forces on the rocket throughout its flight. Figure 6 illustrates the forces acting on the rocket up to burnout. It should be noted that Scenario 1 indicates these results correspond to a launch angle of 5 degrees and a payload of 226.8 kg.

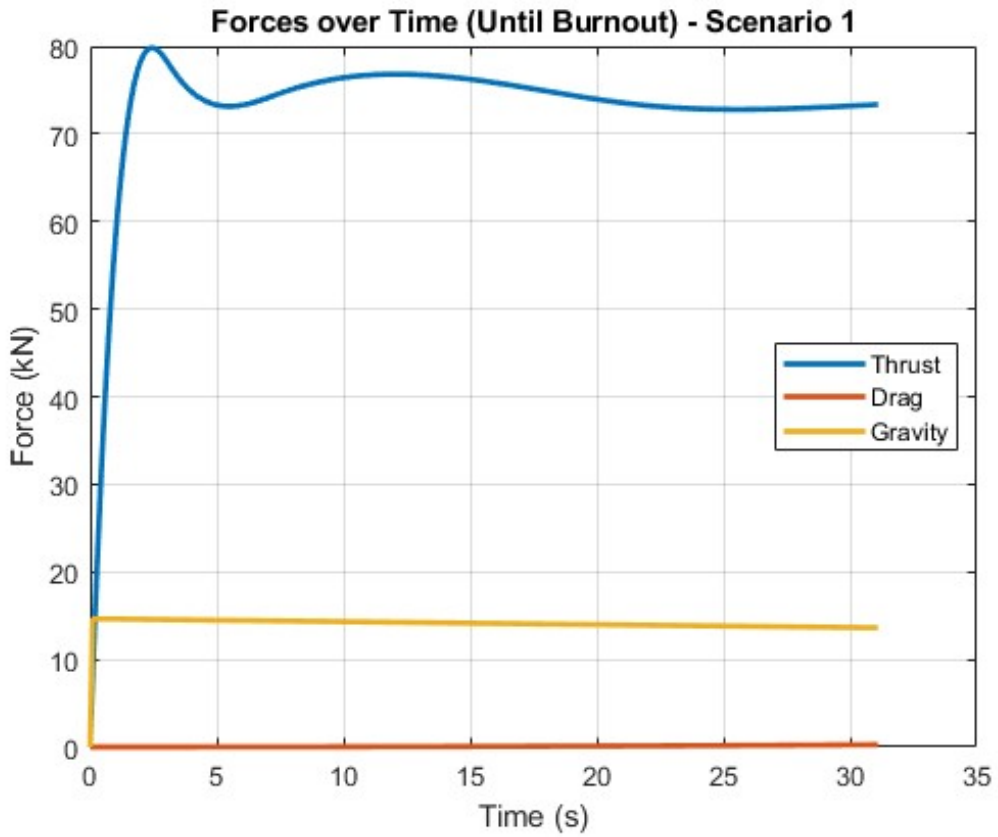


Figure 6: Forces Until Burnout

The thrust spikes initially until it peaks at nearly 80 kN, and the force of drag is near zero for this portion of the flight, while the rocket remains in at low altitudes it experiences higher density, however, the rocket is still traveling at a slow enough speed that the drag is negligible compared to the thrust and the force of gravity. The force of gravity can decrease throughout the burn time, as the rocket expels fuel, its mass decreases, and the force of gravity it experiences follows suit.

Knowing that the drag is a function of the velocity squared, we can inspect the velocity of the rocket throughout its flight to see when we might expect the drag to peak. This is shown in the figure below.

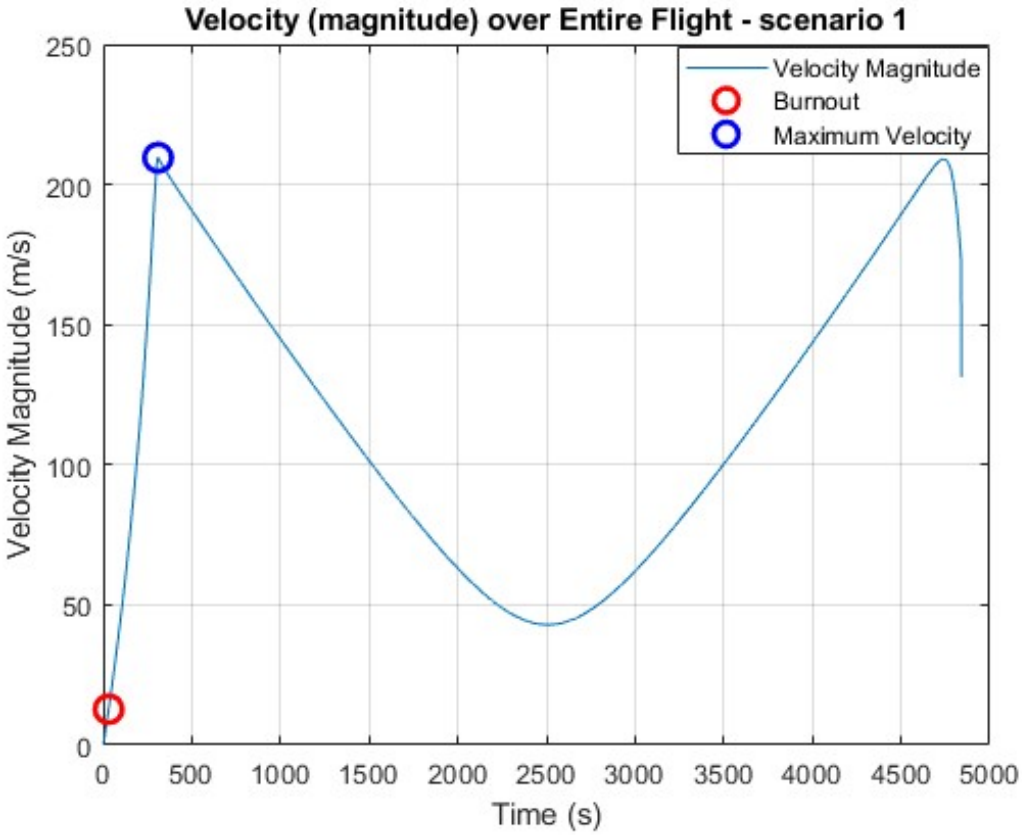


Figure 7: Velocity - Entire Flight Scenario 1

This Figure shows the velocity throughout the flight, and the velocity at burnout as well as the rocket's peak velocity. We can compare the points to the forces experienced by the rocket throughout the flight shown below.

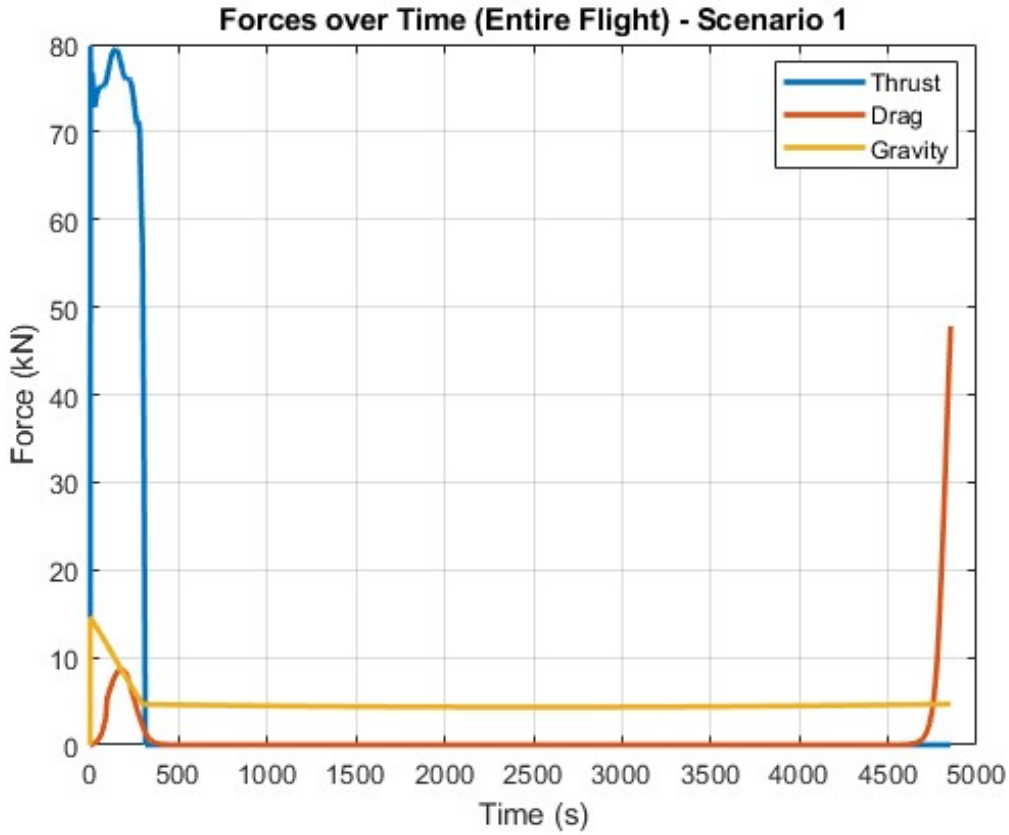


Figure 8: Forces Over Entire Flight

We observed that the drag, as anticipated, increases with the rocket's peak velocity. However, drag is also influenced by density, which decreases as the peak velocity is approached. Moreover, the drag coefficient is associated with the Mach number at which we are traveling. At approximately 2500 seconds, the rocket reaches its apogee, where all velocity is directed horizontally. As the rocket enters Earth's atmosphere and accelerates, the drag force begins to rise again, reaching its maximum just before impact.

The gravitational force acting on the rocket diminishes rapidly as fuel is expended until burnout. Subsequently, as the rocket ascends, the force of gravity continues to decrease. This reduction is due to the atmospheric model accounting for the decrease in gravitational acceleration as a function of altitude. Indeed, the gravitational force experienced by the rocket is minimal at apogee. Beyond this point, during the rocket's descent, the acceleration due to gravity begins to increase again.

References

References

[1] Brent Lewis (2024). *Complete 1976 Standard Atmosphere*, MATLAB Central File Exchange. Available: https://www.mathworks.com/matlabcentral/fileexchange/13635-complete-1976-standard-atmosphere. Retrieved April 9, 2024.

[2] J. Etele, *Fundamentals of Transatmospheric & Space Propulsion*. The Aldanox Group, Inc., 2022.

[3] Etele, J. (2024). *Trajectory assignment V2*. Carleton University.

[4] *Black Brant V Launch Vehicle*, [Online Document], Available: https://sites.wff.nasa.gov/code810/vehicles/Black$_{B}rant_{V}.pdf.Accessed : March17, 2024.